# Yuning Wu

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## RESEARCH INTEREST

**Advancing Adaptive, Ethical, and Context-Aware AI-Driven Robots for Human-Centered Collaboration**
Human-robot interaction, human-centered AI, robot learning, machine learning, computational design

## EDUCATION

2019.8 - 2024.8 **Carnegie Mellon University**, Pittsburgh, USA
Ph.D. in Computational Design
*Dissertation: "A Human-Centered, Reinforcement Learning Driven Robotic Framework for Assisting Construction Workers"*
Co-Advisors: Daniel Cardoso Llach (Computational Design), Jean Oh (Robotics)

2019.8 - 2021.5 **Carnegie Mellon University**, Pittsburgh, USA
M.S. in Machine Learning
Research Advisor: Katerina Fragkiadaki

2010.8 - 2017.7 **Tsinghua University**, Beijing, China
M.Arch. & B.Arch. in Architecture
Focused on how robots can artistically build and interact with the physical space

## RESEARCH HIGHLIGHTS

### A Reinforcement Learning Driven "Work Companion Rover" for Assisting Construction Workers

**Husky** is an RL-driven "work companion rover" built for contextually assisting carpentry workers on-site. By bearing heavy loads and semi-autonomously navigating complex, dynamic construction environments, Husky accompanies workers by delivering and carrying essential tools and materials in the active workflow, reducing physical strain on workers and allowing them to focus on skilled facets of their work.

- Design and architect a multi-sensor robot prototype from a basic chassis (scratch).
- Leverage SoTA RL-based social robot navigation methods to enhance comfortable worker-robot interactions in complex indoor construction workspaces.
- Propose a novel approach to contextually align a generically pre-trained RL model to context-specific worker behavioral patterns while effectively mitigating policy shift in Sim2Real.
- Utilize SoTA LiDAR Odometry and Mapping (LOAM) methods for unstructured site mapping and robot state estimation.
- Customize an efficient multimodal stack (LiDAR + multiple cameras) to robustly detect and track workers' movements in obstructive and cluttered construction environments.

**A Distributed, Scalable RL-Driven Multi-Drone Framework for Flexible Task Cooperation on Complex Geometric Structures**

**Pixhawks** is a fleet of drones driven by an RL-based multi–agent collision avoidance algorithm for conducting flexible tasks such as brick-laying and spray-coating. The algorithmic framework enables a variable number of drones to adaptively and collaboratively execute tasks particularly related to complex-shaped geometric structures, showcasing a robotic framework for handling intricate real-world designs.

- Develop a scalable, decentralized, RL-based multi-agent collision avoidance algorithm to ensure efficient multi-drone operation within limited space.
- Develop a multi-drone simulation environment using Unity and MLAgents.
- Customize a generalizable, curriculum-based Sim2Real pipeline using Unity MLAgents to bridge virtual training and real-world application.
- Conduct a proof-of-concept experiment with a physical drone prototype to validate the practicality and effectiveness of the framework.
- Analyze the evolving role of human workers within the emerging technical system.

**Learning Dense Reward for Contact-Rich Manipulation from Limited Human Demonstrations**

**A dense reward** —a continuous metric that reflects task progress— can be efficiently learned, rather than expert engineered, from multimodal sensory data (vision and force/torque) using a limited number of human demonstrations. The trained reward model can be used in RL for contact-rich robotic manipulation tasks such as USB insertion, door opening, and joint assembly.

- Design a novel Temporal Variant Forward Sampling (TVFS) method, robust and cost-efficient in generating learning samples from human demonstrations within a physical simulation environment.
- Introduce a self-supervised latent representation learning architecture that efficiently leverages the temporal relationships within sequential sample pairs generated by TVFS.

## PUBLICATIONS

[1] **Yuning Wu**, Jiaying Wei, Jean Oh, Daniel Cardoso Llach, "*Towards Human-Centered Construction Robotics: An RL-Driven Companion Robot For Contextually Assisting Carpentry Workers*", accepted to IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Abu Dhabi, 2024. Link.

[2] **Yuning Wu***, Emek Erdolu*, Jiaying Wei, Jean Oh, Daniel Cardoso Llach. "*Robot in the Loop: An Ethnography-Based Approach to Robotics and Reinforcement Learning to Support On-Site Construction Work*", under review, submitted to Springer Nature Construction Robotics, Special Issue on AI and Robotics. Under review. Link.

[3] **Yuning Wu**, Jieliang Luo, Hui Li, "*Learning Dense Rewards with Temporal Variant Self-Supervision*", in the IEEE International Conference on Robotics and Automation (ICRA), Reinforcement Learning for Contact-Rich Manipulation Workshop, Philadelphia, 2022. Link.

[4] Zhihao Fang, **Yuning Wu**, Ammar Hassonjee, Ardavan Bidgoli, Daniel Cardoso Llach, "*Towards an Architectural Framework for Distributed, Robotically-Assisted Construction: Using Reinforcement Learning to Support Scalable Multi-Drone Construction in Dynamic Environments*", In Proceedings of the 40th Annual Conference of the Association for Computer-Aided Design in Architecture (ACADIA), Philadelphia, 2020. Link.

[5] Omer Akin, **Yuning Wu**, "*State Space Paradox of Computational Research in Creativity*", in the 19th Heron Island Conference on Computational and Cognitive Models of Creative Design (HI), 2019. Link.

[6] **Yuning Wu**. A Human-Centered, Reinforcement Learning Driven Robotic Framework for Assisting Construction Workers. Ph.D. Dissertation, Carnegie Mellon University, 2024.

## HONORS AND AWARDS

2023 & 2024 **CMU Presidential Fellowship.** Carnegie Mellon University, TCS.

2022 **Manufacturing Futures Initiatives Fellowship.** Carnegie Mellon University.

2022 **Autodesk AI Merit Scholarship.** Autodesk AI Lab.

2021 **Manufacturing Pennsylvania Fellowship.** State of Pennsylvania.

2020 **GSA-Provost GuSH Grant.** Carnegie Mellon University.

2019 **LEGO Scholarship.** LEGO Foundation.

2017 **Distinctive Graduation Thesis.** Tsinghua University.

2011-2014 **Siyuan Exceptional Student Fellowship.** Tsinghua University.

## WORK EXPERIENCE

2020.8 - 2023.12 **Carnegie Mellon University**, Pittsburgh, PA
Graduate Researcher and Technical Lead
Experienced the full research cycle of the "Rethinking Automation in Construction" project, including funding proposal, lab space setup, robot hardware purchase, robot architecture and prototyping, algorithm design, and final evaluation among actual workers.

2021.5 - 2022.5 **Autodesk AI Lab & Robotics Lab**, San Francisco, CA
Research Intern and Collaborator
Worked on Inverse Reinforcement Learning (IRL) methods for contact-rich manipulation.

2017.12 - 2019.8 **Metascope Technology**, Beijing, China
Co-founder and CTO

## TALKS AND PRESENTATIONS

2023.5 **Cornell Tech, TCS Innovation Forum 2023**. "Towards Humanely Supporting Heavy Manual Labor: Leveraging AI/ML Advances to Develop a 'Work Companion Robot' for Construction Workers"

2023.5 **Bosch Leadership and AI Team Visit and Research Exchange**. "Collaboratively Assist, Rather Than Brutally Replace: A Reinforcement Learning Driven Robotic Framework For Supporting Construction Workers On-Site"

2022.11 **CMU Manufacturing Futures Institute Research Seminar**. "Rethinking Automation in Construction: Robotically Supported Collaborative Work in Heavy Manual Construction"

2022.10 **CMU 48-620 Graduate Seminars: Situating Research**. "Love, Death, and Robots: Designing AI-Driven Robots For Real-World Challenges, Complexities and Nuances"

2022.10 **CMU Board of Trustees Research Exhibit.** "Rethinking Automation in Construction: Introducing Mobile Robots into Existing Real-World Construction Scenes"

2022.5 **ICRA Reinforcement Learning for Contact-Rich Manipulation Workshop.** "Learning Dense Rewards with Temporal Variant Self-Supervision"

2021.11 **CMU 48-770 Graduate Seminars: Inquiries into Machine Learning and Design.** "Love, Death, and Robots: How Reinforcement Learning Driven Robots Can be Trained to Handle Real-World Nuances"

## TEACHING

2022 Spring & Fall — **Collaborative Instructor,** (48-749) Special Topics in Computational Design: Rethinking Automation in Construction, research-based, Carnegie Mellon University. Led and mentored a group of 4-6 MSCD master students in developing their individual research projects in AI and robotics, closely related to the ReAC project.

2021 Spring — **Teaching Assistant,** (10-403) Deep Reinforcement Learning and Control, course-based, Carnegie Mellon University. Engaged in designing hands-on labs, assignments, and exams.

2020 Spring — **Teaching Assistant,** (48-715) MSCD Pre-Thesis I, research-based, Carnegie Mellon University. Supported a group of 10-15 MSCD students in forming their master thesis research through literature review, writing, and discussion.

2020 Fall — **Teaching Assistant,** (48-716) MSCD Pre-Thesis II: Topic Research, research-based, Carnegie Mellon University. Helped the MSCD cohort in developing their thesis proposal through discussions, presentations, and reviews.

2021 Spring — **Teaching Assistant,** (48-718) MSCD Thesis Defense, research-based, Carnegie Mellon University. Supported the same MSCD cohort in preparing the final steps of their thesis defense. Consistently followed through the complete cycle of MSCD thesis research mentoring.

## REFERENCES

**Dr. Daniel Cardoso Llach**

Associate Professor in Computational Design
College of Fine Art
Carnegie Mellon University

**Dr. Jean Oh**

Associate Research Professor in Robotics
Robotics Institute
School of Computer Science
Carnegie Mellon University

**Dr. Jieliang (Rodger) Luo**

Senior Principal Research Scientist
Autodesk AI Lab

**Dr. Hui Li**

Senior Principal Research Scientist
Autodesk Robotics Lab